



SDG 8: DECENT WORK AND ECONOMIC GROWTH

Which individuals are most likely to fall below the median income line:

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Hackathon 5





- **The Problem:** Most workers do not know what similar workers in their job and state earn, so they cannot tell if their pay is unusually low.
- **The Context:** In the US, women working full-time earn 83% of men's median earnings, dropping to 74% in some occupations. A large part of this gap sits within the same job.
- **Our Prediction:** Predicting if a US worker's hourly wage is above the median for their specific occupation and state.

THE USER & ETHICAL REASONING

Target audience: labor rights organizations, career counselors, and individual workers auditing their own fair pay

Model Inputs: Strictly job-related information (occupation, industry, age, education, usual hours, weeks worked, employer type).

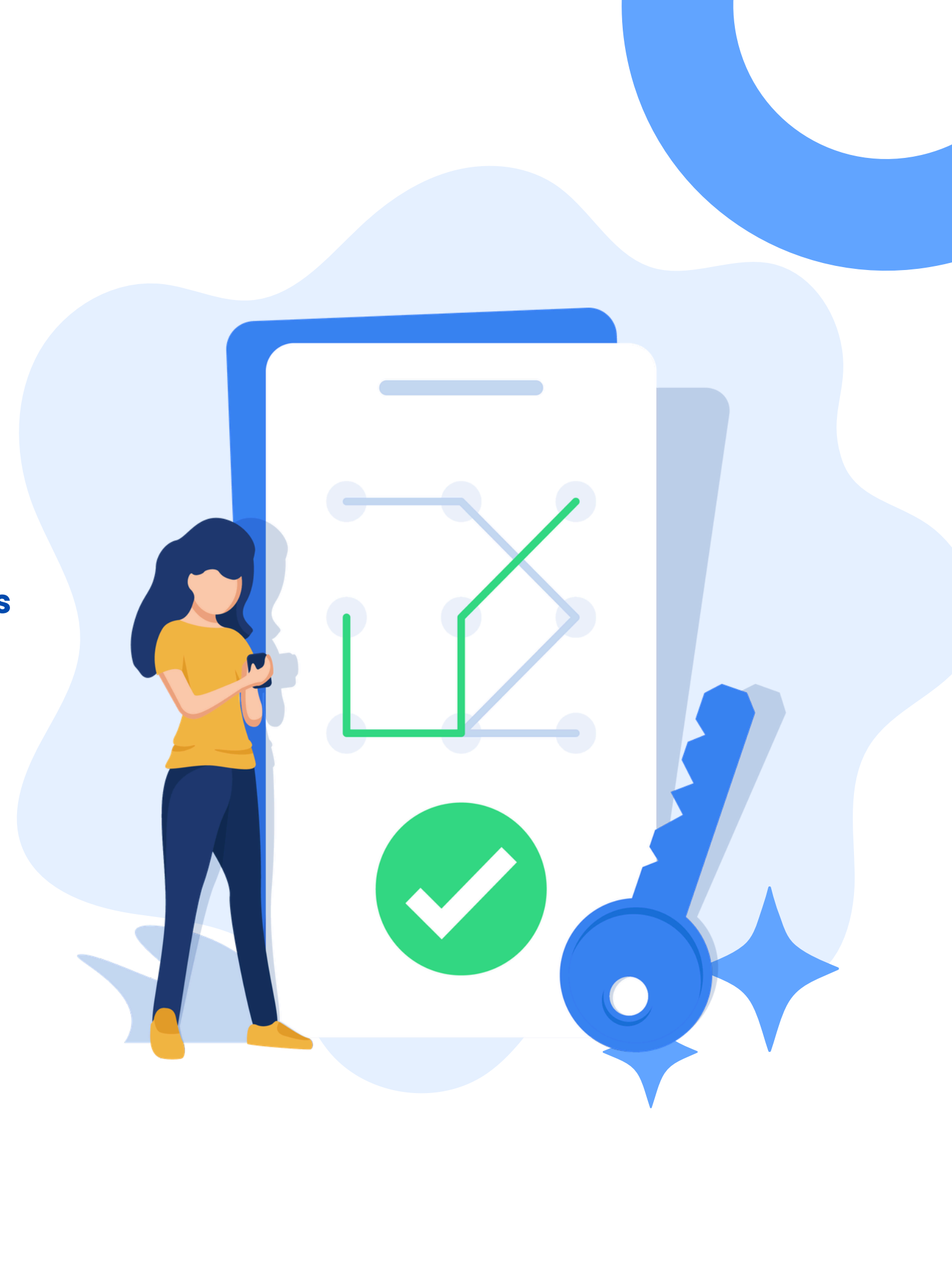
Ethical Firewall: Sex, race, and citizenship are not used as model inputs. If the AI sees gender or race, it might copy human discrimination. We only use those details at the end to audit the model

The Audit: Sensitive demographics are used only to audit the final model for biases.



THE PIPELINE & OUR HARDEST DECISION

- Pipeline: Clean Split → ColumnTransformer (Imputation + Scaling) → GridSearchCV Tuning.
- **The Hardest Decision: Both AI mistakes are dangerous for the worker**
- **Cost of Errors: Both types of errors carry significant real-world risks for the user, making it incredibly difficult to balance the model.**
- **A False Negative (the model says you are fairly paid when you are actually underpaid) means the worker remains exploited. Conversely**
- **False Positive (the model says you are underpaid when you actually have a good salary) could lead a worker to unnecessarily confront their boss or quit a stable job based on bad information.**



ETHICAL REFLECTION

If the model heavily penalizes certain demographics (gender or race), we risk automating historical biases (automation bias).

For our user, a False Negative (failing to identify someone who urgently needs help) is far more damaging than a False Positive (offering support to someone who might not strictly need it).

Who is Left Out: Undocumented workers or individuals operating entirely in the informal economy, as they are not represented in this official dataset.





THANK YOU!